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JEE Main Physics Practice Sheet — DPP-032

Topic: Dynamics of Uniform Circular Motion — Centripetal Force & Applications
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A particle of mass m moves in a horizontal circular path of radius R with a constant speed v. The magnitude of the average acceleration over a quarter of a revolution is:

(A) $\frac{v^2}{R}$
(B) $\frac{2\sqrt{2}v^2}{\pi R}$
(C) $\frac{4v^2}{\pi R}$
(D) $\frac{\sqrt{2}v^2}{\pi R}$

Q2. A conical pendulum consists of a bob of mass m attached to a light string of length L rotating in a horizontal circle of radius r. If the string makes an angle θ with the vertical, the time period of revolution is:

(A) $2\pi\sqrt{\frac{L\cos\theta}{g}}$
(B) $2\pi\sqrt{\frac{L}{g\cos\theta}}$
(C) $2\pi\sqrt{\frac{L\sin\theta}{g}}$
(D) $2\pi\sqrt{\frac{L}{g}}$

Q3. A car travels along a level circular road of radius $R = 50$ m. If the coefficient of static friction between the tyres and the road is $\mu_s = 0.5$, the maximum safe speed without skidding is ($g = 10$ m/s^2):

(A) 25.0 m/s
(B) 10.0 m/s
(C) 15.8 m/s
(D) 20.0 m/s

Q4. A curved track of radius R is banked at an angle θ. The optimum speed v_0 at which a vehicle can round the curve without relying on friction between tyres and the road is:

(A) $\sqrt{\frac{Rg}{\tan\theta}}$
(B) $\sqrt{Rg\sin\theta}$
(C) $\sqrt{Rg\cos\theta}$
(D) $\sqrt{Rg\tan\theta}$

Q5. A motorcyclist loops inside a cylindrical vertical rotor ('Death Well') of radius $R = 4$ m. If the coefficient of static friction between the wall and tyres is $\mu_s = 0.4$, the minimum horizontal circular speed needed to keep the rider from sliding down is ($g = 10$ m/s^2):

(A) 10 m/s
(B) 4 m/s
(C) 6.32 m/s
(D) 12.5 m/s

Q6. A small coin is placed at a distance $r = 10$ cm from the center of a horizontal rotating turntable. If $\mu_s = 0.4$, the maximum frequency of rotation (in rev/min) for the coin not to slide off is ($g = 10$ m/s$^2, \pi^2 \approx 10$):

(A) 120 rpm
(B) 60 rpm
(C) 30 rpm
(D) 90 rpm

Q7. An aircraft executes a horizontal turn of radius $R = 1000$ m at a uniform speed of $v = 200$ m/s. The angle of banking θ that the wings must make with the horizontal to prevent side-slipping is ($g = 10$ m/s^2):

(A) $\tan^{-1}(2)$
(B) $\tan^{-1}(4)$
(C) $\tan^{-1}(0.25)$
(D) $\tan^{-1}(0.5)$

Q8. A particle of mass m is suspended from the ceiling of a railway carriage by a string of length l. If the train rounds a horizontal circular curve of radius R with speed v, the tension in the string is:

(A) $m\sqrt{g^2 + \frac{v^4}{R^2}}$
(B) $mg - \frac{mv^2}{R}$
(C) $mg + \frac{mv^2}{R}$
(D) $m\sqrt{g^2 - \frac{v^4}{R^2}}$

Q9. In a banked circular road of radius R, angle of banking θ, and coefficient of friction $\mu_s < \tan\theta$, the minimum speed $v_{\min}$ below which a vehicle tends to slip down the bank is:

(A) $\sqrt{Rg\left[\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right]}$
(B) $\sqrt{Rg\left[\frac{\tan\theta + \mu_s}{1 - \mu_s\tan\theta}\right]}$
(C) $\sqrt{Rg(\tan\theta - \mu_s)}$
(D) $\sqrt{Rg\left[\frac{1 - \mu_s\tan\theta}{\tan\theta + \mu_s}\right]}$

Q10. A smooth hemispherical bowl of radius R is rotated about its vertical axis of symmetry with angular speed ω. A small particle placed on its inner surface stays in circular equilibrium at a depth h below the rim ($h < R$). The angular velocity ω is:

(A) $\sqrt{\dfrac{g}{R-h}}$

(B) $\sqrt{\dfrac{g}{h}}$

(C) $\sqrt{\dfrac{g}{R}}$

(D) $\sqrt{\dfrac{g}{\sqrt{R^2-h^2}}}$

Q11. A light string of length L can withstand a maximum tension $T_{\max} = 100$ N. A stone of mass $m = 0.5$ kg is tied to it and whirled in a horizontal circle on a frictionless horizontal tabletop. The maximum speed the stone can have without breaking the string is ($L = 2$ m):

(A) 14.14 m/s

(B) 10.0 m/s

(C) 20.0 m/s

(D) 25.0 m/s

Q12. Two identical bodies A and B, each of mass m, are attached to a central pivot by two light strings of equal length L along the same radial line and rotated on a smooth horizontal table at uniform angular speed ω. The ratio of tension in string section OA to string section AB is:

(A) $1:2$

(B) $3:2$

(C) $2:1$

(D) $3:1$

Q13. A car rounds an unbanked circular curve of radius R with speed v. The distance between its two inner and outer wheels is $2d$, and the height of its center of mass above the ground is h. The maximum speed to avoid overturning is:

(A) $\sqrt{\dfrac{Rgh}{d}}$

(B) $\sqrt{\dfrac{Rgd}{h}}$

(C) $\sqrt{Rgdh}$

(D) $\sqrt{\dfrac{2Rgd}{h}}$

Q14. When a particle moves with uniform speed in a circle of radius R, the work done by the centripetal force over a half revolution is:

(A) Zero

(B) πmv^2

(C) $\dfrac{1}{2}mv^2$

(D) $2mv^2$

Q15. A cyclist bends inwards while negotiating a sharp level horizontal curve of radius R with speed v. The angle θ made by the cycle frame with the vertical is:

(A) $\sin^{-1}\left(\dfrac{v^2}{Rg}\right)$

(B) $\cos^{-1}\left(\dfrac{v^2}{Rg}\right)$

(C) $\tan^{-1}\left(\dfrac{v^2}{Rg}\right)$

(D) $\tan^{-1}\left(\dfrac{Rg}{v^2}\right)$

Q16. A hollow vertical cone of semi-vertical angle α rotates about its axis with constant angular velocity ω. A small block is in circular motion at distance r from the axis on the smooth inner surface. The required angular velocity ω is:

(A) $\sqrt{\dfrac{g}{r\tan\alpha}}$

(B) $\sqrt{\dfrac{g\tan\alpha}{r}}$

(C) $\sqrt{\dfrac{g}{r\sin\alpha}}$

(D) $\sqrt{\dfrac{g\sin\alpha}{r}}$

Q17. A bead of mass m is threaded on a smooth horizontal rod of length L pivoted at one end. A spring of stiffness k and natural length l_0 connects the bead to the pivot. If the rod rotates with constant angular speed $\omega < \sqrt{k/m}$, the stretch x in the spring is:

(A) $\dfrac{m\omega^2 l_0}{k - m\omega^2}$

(B) $\dfrac{kl_0}{k - m\omega^2}$

(C) $\dfrac{m\omega^2 l_0}{k + m\omega^2}$

(D) $\dfrac{kl_0}{m\omega^2}$

Q18. A railway engine of mass M moves at speed v along a curved track of radius R. The outer rail is raised above the inner rail by height h (gauge separation d). For no lateral thrust on the rails:

(A) $h = \dfrac{v^2 d}{Rg}$

(B) $h = \dfrac{Rg}{v^2 d}$

(C) $h = \dfrac{v^2}{Rgd}$

(D) $h = \dfrac{vd}{Rg}$

Q19. An open container filled with liquid of density ρ is rotated at constant angular velocity ω about its central vertical axis. The shape of the free surface of the liquid is:

(A) Hemispherical

(B) Hyperbolic

(C) Parabolic, $y = \dfrac{\omega^2 r^2}{2g}$

(D) Elliptical

Q20. A tube of length L filled completely with an incompressible liquid of total mass M is closed at both ends and rotated horizontally about one end with angular speed ω. The force exerted by the liquid on the outer closed end is:

(A) $M\omega^2 L$

(B) $\dfrac{1}{2}M\omega^2 L$

(C) $\dfrac{1}{3}M\omega^2 L$

(D) $\dfrac{1}{4}M\omega^2 L$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-032)

Dynamics of Uniform Circular Motion — Centripetal Force & Applications

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	C	16	A
2	A	7	B	12	B	17	A
3	C	8	A	13	B	18	A
4	D	9	A	14	A	19	C
5	A	10	A	15	C	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (B)
Initial velocity $\vec{v}_i = v\hat{j}$ and after quarter revolution $\vec{v}_f = -v\hat{i}$.
Change in velocity:

$$|\Delta\vec{v}| = |(-v\hat{i}) - (v\hat{j})| = \sqrt{v^2 + v^2} = \sqrt{2}v$$

Time taken for a quarter revolution ($\theta = \pi/2$):

$$\Delta t = \frac{\frac{\pi}{2}R}{v} = \frac{\pi R}{2v}$$

Magnitude of average acceleration:

$$a_{\text{avg}} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{\sqrt{2}v}{\frac{\pi R}{2v}} = \frac{2\sqrt{2}v^2}{\pi R}$$

Sol 2. Option (A)
For the conical pendulum: Vertical equilibrium: $T\cos\theta = mg \implies T = \frac{mg}{\cos\theta}$.
Horizontal centripetal force:

$$T\sin\theta = m\omega^2 r = m\omega^2(L\sin\theta)$$

$$\frac{mg}{\cos\theta}\sin\theta = m\omega^2 L\sin\theta \implies \omega = \sqrt{\frac{g}{L\cos\theta}}$$

Time period:

$$T_p = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

Sol 3. Option (C)
On a level unbanked curve, static friction provides the required centripetal force:

$$f_s = \frac{mv^2}{R} \le \mu_s N = \mu_s mg \implies v \le \sqrt{\mu_s Rg}$$

$$v_{\max} = \sqrt{0.5 \times 50 \times 10} = \sqrt{250} \approx 15.81 \text{ m/s}$$

Sol 4. Option (D)
For a smooth banked curve, the horizontal component of the normal reaction provides the centripetal force: $N\sin\theta = \frac{mv_0^2}{R}$ and $N\cos\theta = mg$.
Dividing equations:

$$\tan\theta = \frac{v_0^2}{Rg} \implies v_0 = \sqrt{Rg\tan\theta}$$

Sol 5. Option (A)
The cylindrical wall provides the normal force: $N = \frac{mv^2}{R}$.
Vertical equilibrium requires static friction to balance gravity:

$$f_s = mg \le \mu_s N = \mu_s\frac{mv^2}{R} \implies v \ge \sqrt{\frac{Rg}{\mu_s}}$$

$$v_{\min} = \sqrt{\frac{4 \times 10}{0.4}} = \sqrt{100} = 10 \text{ m/s}$$

Sol 6. Option (B)
For the coin to stay on the turntable without slipping:

$$m\omega^2 r \le \mu_s mg \implies \omega \le \sqrt{\frac{\mu_s g}{r}} = \sqrt{\frac{0.4 \times 10}{0.1}} = \sqrt{40} \text{ rad/s}$$

Using $\omega = 2\pi f$:

$$f = \frac{\sqrt{40}}{2\pi} \approx \frac{2\sqrt{10}}{2\sqrt{10}} = 1 \text{ rev/s} = 60 \text{ rpm}$$

Sol 7. Option (B)
For the aircraft in a coordinated horizontal banked turn, the lift force L provides: $L\sin\theta = \frac{mv^2}{R}$ and $L\cos\theta = mg$.

$$\tan\theta = \frac{v^2}{Rg} = \frac{200^2}{1000 \times 10} = \frac{40000}{10000} = 4 \implies \theta = \tan^{-1}(4)$$

Sol 8. Option (A)
The string deflects outward by an angle θ such that: $T\cos\theta = mg$ and $T\sin\theta = \frac{mv^2}{R}$.
Squaring and adding both equations:

$$T^2 = (mg)^2 + \left(\frac{mv^2}{R}\right)^2 \implies T = m\sqrt{g^2 + \frac{v^4}{R^2}}$$

Sol 9. Option (A)
At minimum speed, friction acts upwards along the banked slope: $N\cos\theta + f\sin\theta = mg$ and $N\sin\theta - f\cos\theta = \frac{mv^2}{R}$ with $f = \mu_s N$.

$$\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta} = \frac{v^2}{Rg} \implies v_{\min} = \sqrt{Rg\left[\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right]}$$

Sol 10. Option (A)
Radius of horizontal circle: $r = \sqrt{R^2 - (R-h)^2}$.
If θ is angle with vertical: $\cos\theta = \frac{R-h}{R}$ and $\sin\theta = \frac{r}{R}$.
From equilibrium: $N\cos\theta = mg$ and $N\sin\theta = m\omega^2 r$.

$$\tan\theta = \frac{\omega^2 r}{g} \implies \frac{\sin\theta}{\cos\theta} = \frac{\omega^2(R\sin\theta)}{g} \implies \omega = \sqrt{\frac{g}{R\cos\theta}} =$$

Sol 11. Option (C)
Centripetal force is provided entirely by string tension:

$$T = \frac{mv^2}{L} \le T_{\max} \implies v_{\max} = \sqrt{\frac{T_{\max}L}{m}} = \sqrt{\frac{100 \times 2}{0.5}} = \sqrt{400} = 20 \text{ m/s}$$

Sol 12. Option (B)
For outer mass B (at distance $2L$):

$$T_{AB} = m\omega^2(2L) = 2m\omega^2 L$$

For inner mass A (at distance L):

$$T_{OA} - T_{AB} = m\omega^2 L \implies T_{OA} = T_{AB} + m\omega^2 L = 2m\omega^2 L + m\omega^2 L = 3m\omega^2 L$$

Ratio:

$$\frac{T_{OA}}{T_{AB}} = \frac{3m\omega^2 L}{2m\omega^2 L} = \frac{3}{2}$$

Sol 13. Option (B)
At the verge of toppling over, the normal force on the inner wheels drops to zero ($N_{\text{inner}} = 0$).
Taking torque about the center of mass:

$$N_{\text{outer}} d - fh = 0 \implies (mg)d - \left(\frac{mv^2}{R}\right) h = 0 \implies v_{\max} = \sqrt{\frac{Rgd}{h}}$$

Sol 14. Option (A)
Centripetal force is always perpendicular to instantaneous displacement ($\vec{F}_c \cdot d\vec{r} = 0$). Hence, work done by centripetal force is identically zero over any interval.

Sol 15. Option (C)
Normal force N and ground friction f_s pass through the center of gravity of the cyclist when leaning at angle θ to the vertical:

$$\tan\theta = \frac{f_s}{N} = \frac{mv^2/R}{mg} = \frac{v^2}{Rg} \implies \theta = \tan^{-1}\left(\frac{v^2}{Rg}\right)$$

Sol 16. Option (A)
Normal force N is perpendicular to the cone wall (at angle α to horizontal).
$N\cos\alpha = mg$ and $N\sin\alpha = m\omega^2 r$.

$$\tan\alpha = \frac{\omega^2 r}{g} \implies \omega = \sqrt{\frac{g\tan\alpha}{r}} = \sqrt{\frac{g}{r\cot\alpha}}$$

When α is the semi-vertical angle, normal reaction is inclined at α to vertical:

$$N\sin\alpha = mg, \quad N\cos\alpha = m\omega^2 r \implies \omega = \sqrt{\frac{g}{r\tan\alpha}}$$

Sol 17. Option (A)
Total distance of bead from pivot is $(l_0 + x)$.
The inward spring force provides centripetal acceleration:

$$kx = m\omega^2(l_0 + x) \implies kx - m\omega^2 x = m\omega^2 l_0 \implies x = \frac{m\omega^2 l_0}{k - m\omega^2}$$

Sol 18. Option (A)
For track banking angle $\theta \approx \sin\theta = h/d$:

$$\tan\theta \approx \frac{h}{d} = \frac{v^2}{Rg} \implies h = \frac{v^2 d}{Rg}$$

Sol 19. Option (C)
At any point on the free surface at radial distance r, the tangent slope satisfies:

$$\frac{dy}{dr} = \frac{\omega^2 r}{g} \implies y = \int \frac{\omega^2 r}{g}\, dr = \frac{\omega^2 r^2}{2g}$$

The surface profile is a paraboloid of revolution.

Sol 20. Option (B)
Consider a differential element of length dr and mass $dm = \frac{M}{L}dr$ at distance r.
Centripetal force required by the liquid column:

$$F = \int_0^L (dm)\omega^2 r = \frac{M}{L}\omega^2 \int_0^L r\, dr = \frac{M}{L}\omega^2\left(\frac{L^2}{2}\right) = \frac{1}{2}M\omega^2 L$$